

Solution Architecture

Date	30 September 2026	Team ID	6ab22029e880316a482c5f2a
Project Name	EduGenie – AI-Powered Educational Assistant	GitHub	https://github.com/theopintheo/EduGenie
Team Members	Theopin B, Antony Ragul A, Saravanan S, Karthick M	Status	Completed / working implementation

Layer	Role	Technology
Presentation	Learner tabs, input, results and quiz interaction	HTML, CSS, JavaScript, Jinja2
API	Routes, validation, response formatting, health check	FastAPI + Pydantic
Feature Logic	Independent prompt/feature implementations	qna.py, explanation_module.py, summary_module.py, quiz_module.py, learning_path.py
AI Service	Shared model client, configuration, fallback	google-genai + AIService
External AI	Generative inference	Configured Gemini/Antigravity model
Configuration	Secrets and model settings	python-dotenv + .env

Architecture Flow

Browser UI ■ FastAPI Router ■ Feature Module ■ AIService ■ Gemini API. The AIService can return deterministic demo responses when no API key is configured.

The current project is intentionally modular and monolithic rather than microservice-based, keeping deployment and maintenance simple for the current scope.